# AI-Based Expense Tracker with Analytics and Investment Recommendations

M. Prem Kumar
*Dept. of AI & ML*
*VISTAS, Chennai, Tamil Nadu, India*

A. Subash
*Dept. of AI & ML*
*VISTAS, Chennai, Tamil Nadu, India*

V. Dhanush Raj
*Dept. of AI & ML*
*VISTAS, Chennai, Tamil Nadu, India*

*Project Guide: Dr. Meenakshi N., Department of Artificial Intelligence and Machine Learning, Vels Institute of Science, Technology & Advanced Studies (VISTAS), Chennai, India*

***Abstract*—Managing personal expenses well is key to financial stability, yet many people struggle to keep track of daily spending, understand where their money goes, and plan their savings. This paper presents an AI-based expense tracker with analytics and investment recommendations, a web-based system in which users record income and expenses, categorize transactions, and monitor budgets on an interactive dashboard. Machine learning is applied to past transaction data to identify spending patterns, predict future expenses, and flag overspending. Users can also define financial goals, such as buying a car or funding an education plan, and receive investment suggestions (Fixed Deposits, Mutual Funds, Gold, or Stocks) matched to their income, savings, goal amount, and risk preference. The application uses React.js for the frontend, Flask for the backend, SQLite for persistent storage, and scikit-learn for the analytics engine. By bringing expense tracking, predictive analytics, budget alerts, and goal-based investment guidance together on one platform, the system aims to improve financial literacy and encourage more disciplined personal money management.**

***Index Terms*—artificial intelligence, machine learning, expense tracking, financial analytics, expense prediction, investment recommendation, budget management, goal-based saving.**

## I. Introduction

Keeping personal finances in order helps people rein in spending, build the habit of saving, and work toward bigger long-term goals. Day-to-day transactions pile up quickly across food, transport, shopping, education, and bills, so recording and making sense of them by hand takes a lot of time and invites mistakes. Notebooks and spreadsheets leave all the arithmetic and pattern-spotting to the user, and many of today's expense apps stop at storing records, offering little help in understanding how a person actually spends or what they will need next.

In this paper we propose an AI-based expense tracker with analytics and investment recommendations. It is a web application where users enter their income and expenses by hand, assign each transaction a category, and see everything summarized on one dashboard. A machine-learning layer studies the recorded data to find spending patterns, estimate upcoming expenses, and raise overspending alerts. Users can also set specific financial goals and get a suggested investment category based on their income, expenses, savings, goal amount, and risk preference, which takes the tool beyond plain bookkeeping and toward planning around goals.

### *A. Objectives*

The system is designed to meet the following objectives:

- To build an easy-to-use tool for tracking income and expenses.
- To let users enter and categorize their own transactions.
- To study past data and bring out useful spending insights.
- To forecast future expenses from earlier spending habits.
- To warn users when a budget limit is crossed.
- To help users set financial goals and follow their progress.
- To suggest suitable investments using income, savings, goal, and risk preference.

## II. Literature Review

Many researchers have looked at how AI and machine learning can support personal finance. Table I lists five recent studies that shaped our design, along with the techniques they used, what they contributed, and the limitations they reported.

Chahar *et al.* developed an AI-based personal finance assistant that uses machine learning and natural language processing to produce customized budget advice and financial education material, and tested it with an early prototype [1]. Jadhav and Patil presented a hybrid machine-learning framework that merges multiple classifiers for categorizing expenses and building budgets; it reached a recommendation accuracy of about 98.5% but is harder to interpret [2]. Kumar *et al.* used Long Short-Term Memory (LSTM) networks on historical time-series data to forecast income, expenses, and savings, obtaining around 85% accuracy, although the model handled irregular, one-off financial events poorly [3]. Thakur and Jadhav tested a range of regression and ensemble models on a household transactions dataset and found that a voting ensemble regressor performed best, with an $R^2$ of 78.11% and the smallest relative absolute error [4]. Saranya *et al.* paired OCR-based receipt scanning with BERT-based transaction classification and LSTM forecasting, achieving categorization accuracy above 94%, but their approach needs ongoing access to bank, card, and wallet data [5].

TABLE I

SUMMARY OF RELATED WORK

| Ref. | Study (Author, Year) | AI/ML Technique Used | Key Contribution | Reported Limitation |
|---|---|---|---|---|
| [1] | Chahar *et al.*, 2025 | ML/NLP-based predictive analytics, recommendation models | AI assistant offering customized budget advice and financial education | Performs poorly on irregular or one-off expenses |
| [2] | Jadhav and Patil, 2025 | Hybrid ML (Decision Tree, SVM, Random Forest, ANN) | Automatic expense categorization and budgeting, ~98.5% recommendation accuracy | Needs large labeled datasets; hybrid model is hard to explain |
| [3] | Kumar *et al.*, 2024 | LSTM (recurrent neural network) | Predicts income, expenses, and savings from past time-series data | Weak on irregular one-time income or expenses (bonuses, emergencies) |
| [4] | Thakur and Jadhav, 2025 | XGBoost, Random Forest, SVM, MLP, Voting Ensemble | Voting ensemble regressor for future expenses ($R^2$ = 78.11%) | Relies on a generic household dataset rather than per-user data |
| [5] | Saranya *et al.*, 2026 | OCR + BERT (classification) + LSTM (forecasting) | Automated receipt scanning and categorization (>94% accuracy, ±6% forecast variance) | Needs OCR infrastructure and access to sensitive bank/wallet data |

## III. Research Gap

The reviewed studies leave several gaps that motivate this work:

- Several systems depend on direct bank or card integration, which raises privacy and security concerns for users who would rather keep their financial data under their own manual, offline-first control.
- Recommendations are often based on spending history alone, without weighing income, savings, a particular goal amount, and the user's risk appetite together.
- Where goal features exist, they usually just track progress and do not convert the goal into a concrete monthly saving or investment plan.
- Models trained on generic or simulated household datasets may not fit an individual's irregular income and spending.
- Few systems separate essential everyday spending from unnecessary or excessive spending in a way users can act on right away.

Our system tackles these gaps with a manual-entry-first design that does not require bank integration, together with an analytics engine that considers income, expenses, savings, goal amount, and risk preference at once to give goal-based investment suggestions alongside regular budget monitoring.

## IV. Proposed System

The proposed system is a web application in which users enter income and expense transactions by hand, each with an amount, category, description, and date. The backend checks every entry before saving it, and the analytics engine then refreshes the spending summaries, forecasts, and budget status. Users can also define a financial goal (a target amount and time frame); the system checks whether it is achievable and suggests an investment category.

### *A. System Architecture*

Fig. 1 shows the modular architecture. The User Data Input Module collects the income, expenses, goals, and preferences that the user enters manually. The Frontend Interface Module, built in React, contains the login/sign-up flow, the dashboard, the income and expense screens, analytics views, the goal planner, and the investment-suggestion display. The Core Backend Processing Module, written in Flask, handles authentication, income and expense processing, goal management, financial calculations, and API management. Data is saved in the Database Storage Module (SQLite), which holds users, income, expenses, goals, bills, and preferences. The AI/ML Engine Module works on this stored data for spending-pattern analysis, expense prediction, budget recommendation, goal-feasibility analysis, and investment-category suggestion, and the Results and Insights Delivery Module presents its output as dashboards, analytics, and recommendations. A Security and Data Protection layer, which includes user authentication, role-based access control, data encryption, secure API access, and regular backups, cuts across all modules.

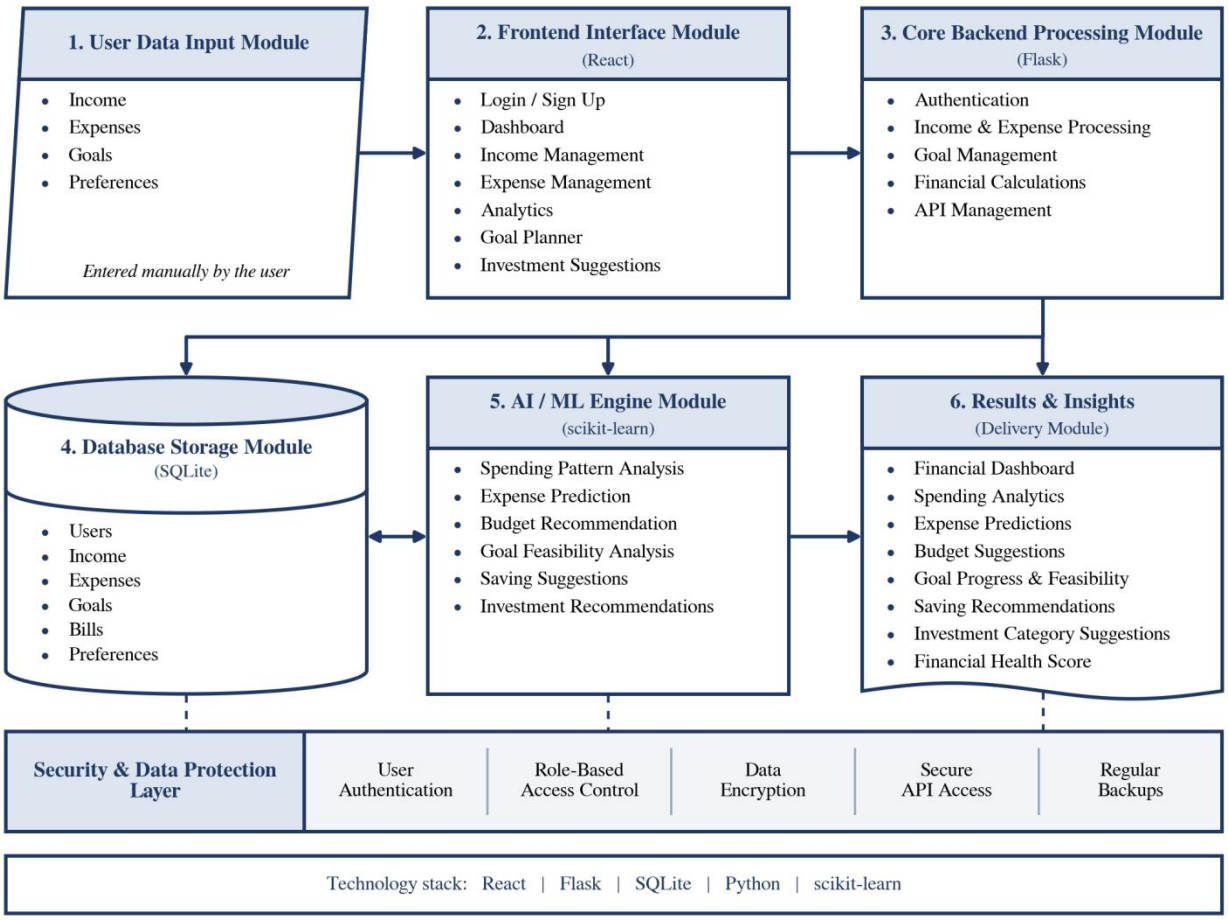



Fig. 1. System architecture of the AI-based expense tracker.

### *B. System Workflow*

Fig. 2 shows the workflow followed for each income or expense entry. Once the user logs in and submits a transaction, the system validates it; an invalid entry is rejected with an on-screen error, while a valid one is saved to the database. The AI/ML engine then categorizes the transaction, updates the spending-pattern analysis, predicts expenses for the near future, and recalculates savings. If the projected expenses in a category go beyond the budget limit the user has set, an overspending alert is raised; if not, the system moves on to check goal feasibility and, where relevant, suggests an investment category. Finally, the dashboard is refreshed with charts, budget status, and savings progress for the user to review.

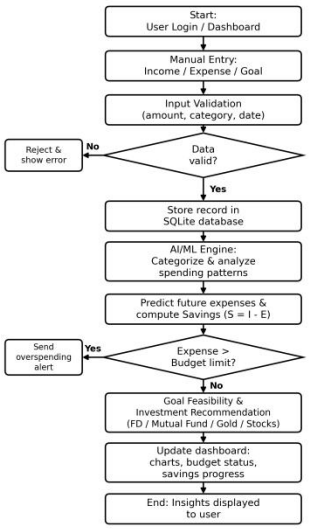



Fig. 2. Expense tracking and recommendation workflow.

## V. METHODOLOGY

### *A. User Authentication*

Registration and login keep each user's financial records tied to their own account. Passwords are stored with a

secure hashing method, and session-based access control decides which pages and API endpoints a logged-in user can reach.

### *B. Income and Expense Entry*

Each transaction is entered through a form that asks for the amount, a category (for example food, transportation, shopping, education, or bills), an optional description, and a date. Common spending areas come as predefined categories, and the design also lets users add their own.

### *C. Data Validation*

Before a transaction is saved, the backend confirms that all required fields are filled in, that the amount is a positive number, and that the chosen category and transaction type are valid. Entries that fail these checks are rejected and the user is asked to correct them, which lowers the chance of corrupted or inconsistent records reaching the database.

### *D. Financial Analytics*

Given a set of income transactions $I_i$ and expense transactions $E_j$ recorded for a user, the system computes three core quantities: total income (1), total expenses (2), and total savings (3).

$$I = \Sigma\, I_i, \quad i = 1, \ldots, n \tag{1}$$

$$E = \Sigma\, E_j, \quad j = 1, \ldots, m \tag{2}$$

$$S = I - E \tag{3}$$

Here, $I_i$ is the $i$-th income transaction, $E_j$ is the $j$-th expense transaction, $n$ is the number of income records, and $m$ is the number of expense records. Together with a category-wise breakdown of $E$, these figures appear on the dashboard so users can see where most of their money goes.

### *E. Budget Monitoring*

For each expense category $c$, the user may optionally set a budget limit $B(c)$. Given the actual expense $E(c)$ logged in that category during the monitoring period, the system applies the rule

$$R(c) = \begin{cases} \text{“Review budget”, if } E(c) > B(c) \\ \text{“Within budget”, if } E(c) \le B(c) \end{cases} \tag{4}$$

This rule marks any category whose spending has crossed the user-defined limit and triggers an overspending alert on the dashboard.

### *F. Goal-Based Investment Recommendation*

Suppose a user sets a financial goal with target amount $G$ and a time horizon of $N$ months, and has current savings of $C_s$, the required monthly saving $M$ is estimated as

$$M = (G - C_s) / N \tag{5}$$

The system compares $M$ against the user’s average historical monthly savings to judge whether the goal is feasible. It then maps the goal’s time horizon and the user’s stated risk preference (low, medium, or high) to an investment category: Fixed Deposits and Gold for short-horizon, low-risk goals; Mutual Funds for medium-term, moderate-risk goals; and Stocks for longer-horizon, higher-risk goals. The result is a concrete, goal-linked starting point for further financial decisions instead of generic investment advice.

## VI. System Implementation

The application follows a client–server architecture. Table II lists the technologies used to build each layer of the system.

TABLE II
TECHNOLOGY STACK

| Component | Technology | Purpose |
|---|---|---|
| Frontend | React.js, HTML, CSS, JavaScript | Login/sign-up, dashboard, transaction forms, analytics and goal-planner views |
| Backend | Python, Flask | Authentication, transaction processing, financial calculations, REST API |
| Database | SQLite | Storage of users, income, expenses, goals, bills, and preferences |
| Analytics engine | Python, scikit-learn | Spending-pattern analysis, expense prediction, recommendation logic |

Development was done in Visual Studio Code, with Node.js and npm for frontend tooling, on a Windows 10/11 machine with at least 4 GB of RAM (8 GB recommended) and 10 GB of free storage. Figs. 3 and 4 show the implemented dashboard and the expense-entry form, respectively.

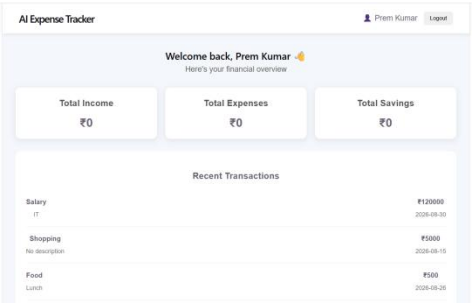



Fig. 3. Dashboard showing income, expenses, savings, and recent transactions.

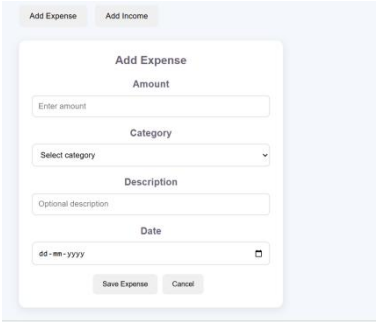



Fig. 4. Expense-entry form with amount, category, description, and date fields.

## VII. Results and Discussion

The application is currently at the first-review stage of implementation, with authentication, the dashboard, and manual transaction entry working. A full quantitative

evaluation of the prediction and recommendation modules is therefore left for the next phase. Evaluation will follow the criteria in Table III, in line with the approach taken in related work [1], [4].

TABLE III
PLANNED EVALUATION CRITERIA

| Evaluation Parameter | Testing Approach |
|---|---|
| Authentication | Test valid and invalid login credentials |
| Transaction validation | Submit valid, incomplete, and invalid transactions |
| Financial calculations | Compare dashboard totals (*I*, *E*, *S*) against manually computed values |
| Expense prediction | Compare predicted and actual expense using MAE and $R^2$, following [3], [4] |
| Budget alerts | Test alert triggering across categories with known over- and under-budget scenarios |
| Investment recommendations | Test output across varied goal, savings, and risk-preference inputs |

No numerical results are claimed at this stage. Measured accuracy, error, and user-satisfaction figures will be reported after model training and user testing are finished.

## VIII. Advantages of the Proposed System

- Centralized, manual-entry-first handling of personal financial records with no mandatory bank integration.
- Automatic calculation of income, expenses, and savings, so users avoid manual arithmetic.
- Validated data entry that cuts down incomplete or invalid records.
- Category-wise spending charts that show the main expense areas.
- Overspending alerts driven by user-defined budget limits.
- Investment category suggestions tied to the user's goal instead of generic advice.
- Lightweight, open-source technology stack (React, Flask, SQLite, scikit-learn).

## IX. Limitations and Future Work

Since transactions are entered by hand, the accuracy of the analytics and recommendations depends on how complete and correct the user's input is; missing entries can skew spending summaries. The budget and goal-feasibility logic is currently rule-based and does not yet use models learned from the user's own history. Planned enhancements include:

- Category-specific expense-prediction models (e.g., regression or LSTM-based, as in [3], [4]) trained on each user's own transaction history.
- Optional OCR-based receipt scanning to cut down manual entry, following the approach in [5].
- Interactive drill-down charts for monthly and category-wise views.
- Goal progress tracking with visual milestones and reminders.
- Stronger authentication (e.g., multi-factor) and encryption of stored financial data.
- Deployment as a hosted, multi-device application.

## X. Conclusion

This paper described an AI-based expense tracker with analytics and investment recommendations that brings manual transaction entry, data validation, financial analytics, budget monitoring, and goal-based investment suggestions together in one web application built on React, Flask, SQLite, and scikit-learn. Unlike earlier systems that depend on bank-account integration or look at spending history alone, our design leaves data entry in the user's hands while reasoning over income, expenses, savings, goals, and risk preference together to give practical, goal-linked guidance. The architecture and workflow presented here lay the groundwork for the prediction and recommendation modules now under development, and evaluation against the criteria in Table III is the next milestone.

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